



Exploiting dynamic spatio-temporal correlations for origin-destination demand prediction

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ABSTRACT

Accurate Origin-Destination (OD) demand prediction is fundamental to intelligent transportation systems (ITS), enabling real-time traffic management, dynamic vehicle dispatch, and efficient resource allocation in urban environments. However, OD demand exhibits complex, dynamic, and highly coupled spatio-temporal patterns that remain challenging for existing models. We propose a novel Dynamic Spatio-Temporal Correlation Network (DSTCN) for OD demand forecasting. DSTCN features three key components: (1) a bidirectional demand trend modeling module (Glstm2D) that jointly learns demand evolution from both origin and destination perspectives; (2) a Transformer-based spatial similarity module (Simformer) to dynamically extract and integrate inter-regional correlations across the OD matrix; and (3) a temporal fusion and modeling module (FF-TM) that combines and processes multi-source spatio-temporal features for next-step prediction. Extensive experiments on three large-scale real-world datasets (NYC-TOD2018, NYC-TOD2019, and HZMetro) demonstrate that DSTCN consistently outperforms state-of-the-art baselines across diverse urban scenarios.

1. Introduction

Modern urban transportation systems—including taxis, ride-hailing vehicles, metro networks, and bus services—face persistent challenges such as supply-demand imbalance, passenger delays, and resource inefficiencies. Accurate origin-destination (OD) demand prediction is essential for Intelligent Transportation Systems (ITS), supporting real-time traffic management, dynamic resource allocation, and large-scale route optimization (Dapeng & Xiao, 2021; Han et al., 2022; Qin et al., 2025; Zhang et al., 2021b). By forecasting the number of passengers traveling from each origin to each destination, OD demand models can reduce travel times, minimize idle trips (Yang et al., 2000), alleviate congestion (Wang et al., 2017), and inform urban planning (Xu et al., 2023a; Zeng et al., 2022).

Unlike OD flow prediction, which tracks real-time flow changes between two locations (Huang et al., 2023; Kumarage et al., 2023; Lv et al., 2024), OD demand prediction estimates the number of passengers departing from an area (origin demand) and arriving at another (destination demand), enabling more actionable and anticipatory ITS decisions.

The OD demand matrix at timestamp t , $X_t \in \mathbb{R}^{N \times N}$, records $X_t(i, j)$ as the number of passengers departing region i at time t to region j . Fig. 1(a) visualizes spatial demand in Manhattan, while Fig. 1(b) shows, for example, $X_t(43, 41) = 75$ denotes 75 passengers traveling from region 43 to 41 at time t .

Currently, relatively few methods directly address the OD demand prediction problem (Zhang et al., 2021b). With the advancement of deep learning, Dapeng and Xiao (2021) introduced a dynamic graph decomposition layer to jointly model graph structure and node representations, but their approach struggled to capture temporal dependencies effectively. Liu et al. (2019) proposed a spatio-temporal network based on two-layer CNNs for spatial context learning; however, multiple layers were required to capture long-range correlations, resulting in increased model complexity.

Despite these efforts, methods that systematically analyze and model the evolving trends of regional OD demand remain limited. Accurate capture of these trends, using both historical patterns and real-time fluctuations, can substantially enhance forecast performance in diverse urban areas. In particular, demand changes tend to exhibit consistent

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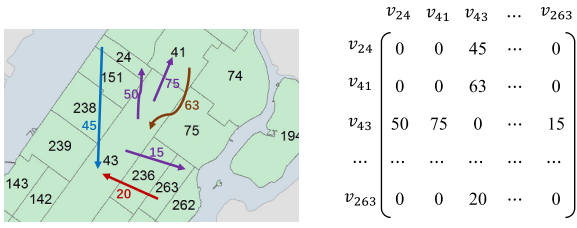


Fig. 1. Example of OD demand at timestamp t . (a) Spatial distribution of crowd demand in Manhattan. (b) Corresponding OD demand matrix X_t .

trajectories from origin to destination regions (see Section 5.9), and effectively leveraging such trends is crucial for improving both the accuracy and adaptability of ITS.

In summary, the OD demand forecasting task presents three main challenges:

- **Network-wide prediction.** OD demand forecasting requires simultaneous estimation of passenger flows from all origins to all destinations across the entire city network.
- **Spatio-temporal tendency correlations.** OD demand exhibits strong spatio-temporal correlations in both the destination and origin directions. Effectively capturing and learning this property poses significant challenges.
- **Temporal similarity among OD regions.** Some grid cells within the OD matrix exhibit dynamic demand similarity. Effectively capturing the demand similarity among all grid cells poses a challenge.

Motivated by these challenges, we propose a Dynamic Spatio-Temporal Correlation Network (DSTCN) for robust and accurate OD demand prediction. The DSTCN framework comprises three major modules: (1) bidirectional demand trend modeling (Glstm2D), (2) a Transformer-based spatial similarity module (Simformer), and (3) a feature fusion and temporal modeling module (FF-TM). Specifically, Glstm2D leverages bidirectional LSTM networks and spatio-temporal graph learning to capture demand evolution from both origin and destination perspectives. The Simformer module utilizes multi-head self-attention to extract OD grid features and to model dynamic relationships based on demand similarity. The FF-TM module then fuses demand trend features from Glstm2D with grid cell features from Simformer and models their temporal evolution.

The main contributions and innovations of this study are summarized as follows:

- We introduce a novel framework for modeling OD demand trends, jointly capturing spatio-temporal correlations between urban regions from both the origin and destination directions. This is crucial for real-time OD demand prediction, dynamic traffic management, and efficient resource allocation in ITS.
- We design a Transformer-based feature embedding module (Simformer) that adaptively integrates external environmental factors into the OD demand matrix. By leveraging the Transformer's capacity for modeling long-range dependencies, this module enhances the responsiveness and predictive accuracy of ITS solutions.
- We propose a multi-head spatio-temporal convolutional layer within the FF-TM module, enabling effective fusion of OD demand features from multiple perspectives and supporting accurate city-wide predictions required for ITS optimization.
- Extensive experiments on three large-scale real-world datasets demonstrate that DSTCN consistently outperforms state-of-the-art OD demand prediction methods, and ablation studies validate the effectiveness of each module.

The remainder of the paper is organized as follows: Section 2 reviews related work; Section 3 defines the problem statement; Section 4 introduces the proposed DSTCN predictive method; Section 5 presents the experimental results, and Section 6 concludes the paper.

2. Related work

2.1. Traffic prediction

Existing traffic prediction methods can be broadly categorized into two groups. The first group consists of traditional statistical and machine learning approaches, such as Auto-Regressive Integrated Moving Average (ARIMA) (Ahmed & Cook, 1979) and Support Vector Regression (SVR) (Drucker et al., 1996). For example, Shekhar and Williams (2007) developed an adaptive parameter estimation methodology for univariate traffic condition forecasting, while Chen et al. (2015) combined SVR with an adaptive genetic algorithm for holiday tourist flow prediction. Although these methods are interpretable and efficient, they are often limited in modeling complex nonlinear dependencies and high-dimensional spatio-temporal data.

The second group comprises deep learning-based approaches that have advanced urban traffic prediction (He et al., 2025; Yang et al., 2025; Zhang et al., 2023b). Zhang et al. (2017) employed a CNN-based deep residual network to model weekly, daily, and recent traffic patterns. Subsequent studies have explicitly modeled spatial dependencies using Graph Convolutional Networks (GCNs). For instance, Zhu et al. (2022) integrated GCNs with knowledge graphs for urban traffic forecasting, while Shin and Yoon (2024) proposed a progressive GCN with causal convolution and gated activation to enhance temporal dynamics. Zhao et al. (2022) introduced a spatio-temporal causal graph attention network with adaptive graph structures to strengthen spatial dependency modeling. Xia et al. (2021) utilized a 3D dynamic GCN for city-wide crowd flow prediction, and Song et al. (2020) designed a spatio-temporal synchronization graph convolution model to capture complex local correlations.

In addition, attention mechanisms have been widely adopted to improve the modeling of spatio-temporal features. Zheng et al. (2020) used attention blocks to capture correlations between sensors and time steps, while Xu et al. (2020) leveraged Transformers to dynamically learn spatial and temporal dependencies. Cheng et al. (2023) proposed a spatio-temporal synchronization graph neural network with attention mechanisms for multi-angle traffic flow prediction. Other works have further enhanced predictive capability by mining spatio-temporal correlations, such as the multi-view localized correlation learning of Gong et al. (2020a), spatio-temporal contrastive learning by Qu et al. (2022), and the attention temporal graph convolutional network by Bai et al. (2021). Beyond these, contrastive frameworks such as GraphST use adversarial adaptation to improve robustness under noisy or incomplete urban graphs (Zhang et al., 2023a). In parallel, spatio-temporal large language models such as UrbanGPT align dependency encoders with instruction-tuned LLMs to enable zero shot generalization across diverse urban tasks (Li et al., 2024).

While these deep learning methods have significantly improved the integration and prediction of spatio-temporal traffic data (Bai et al., 2019; Yao et al., 2018), most urban traffic forecasting studies focus primarily on inflows and outflows at city areas or transportation hubs. This offers only a partial perspective for holistic urban traffic management. In contrast, OD demand prediction seeks to forecast passenger demand between different regions in the transportation network, providing a more comprehensive view of urban mobility and operational needs.

2.2. OD demand prediction

OD demand prediction is a complex task that entails forecasting the number of individuals who depart from an origin or arrive at a destination in specific areas and time intervals. It is important to distin-

guish OD demand prediction from general traffic-flow prediction. While traffic flow prediction typically concerns the movement of vehicles or people along road segments or between regions, OD demand prediction focuses on estimating departures and arrivals for each area. Such granularity is critical for proactive management of taxi dispatch, public transit scheduling, and broader urban mobility services.

Historical averaging (HA) has been a foundational approach for OD demand prediction. Subsequent advances include matrix-decomposition methods (Gong et al., 2020b), which better exploit high-dimensional structure. For example, Feng et al. (2021) proposed a multi-task matrix-factorization model to capture dependencies among zones. In addition, multi-task frameworks that jointly predict crowd flow and OD demand have proven effective. MT-ASTN adopts a shared-private architecture with adversarial and discriminative losses to better disentangle features (Wang et al., 2020), and MBA-STNet introduces a Bayesian component to mitigate data uncertainty (Miao et al., 2023).

Recently, various deep learning methods (Pamuła & Żochowska, 2023; Zhang et al., 2021a, 2022; Zou et al., 2021) have been proposed for OD demand prediction. Representative advances include encoder-decoder architectures for spatio-temporal pattern learning (Ke et al., 2021), integration of geographic and semantic context (Wang et al., 2019), dynamic multi-graph convolutional networks with GANs to model evolving OD topologies (Huang et al., 2022), and multi-resolution decomposition using discrete wavelet transforms (Noursalehi et al., 2021). Adaptive feature fusion frameworks further leverage spatial knowledge graphs and periodic cues (Xu et al., 2023b). The BiST-IF model (Yu et al., 2025) extends this line by introducing bi-directional attention for spatio-temporal dependency modeling and incorporating Out-OD flow to boost prediction accuracy. Contextual fusion is experimentally crucial for traffic prediction (Li et al., 2025). For instance, AEST treats crowd flow and external data as auxiliary tasks, using a hierarchical structure to improve OD prediction on sparse data (Miao et al., 2022). Additionally, for the related task of OD travel time estimation, recent work like DOT (Lin et al., 2023) has employed diffusion models to infer a representative trajectory image from historical data, a technique designed to remove outliers before a final prediction is made.

Despite these advances, most existing methods emphasize global OD demand trends, which may lead models to overlook fine-grained variations. For example, residential areas often exhibit higher origin demand during weekday morning peaks and higher destination demand in the evening, with patterns modulated by proximity to business districts and other factors. To address this, it is essential to model demand trends separately for each area on both the origin and destination sides, leveraging OD periodicities and localized tendencies. Additionally, learning spatial relationships via attention mechanisms can highlight similarities among comparable areas, enabling the capture of detailed local characteristics while maintaining macro-level predictive accuracy. This integrated approach enhances forecasting precision for urban transportation networks.

3. Problem statement and analysis

Before introducing our model, we define the fundamental concepts and notations used throughout this paper, followed by a formal definition of the OD demand prediction problem. Key symbols are summarized in Table 1. We then analyze the intrinsic spatio-temporal correlation patterns in OD demand data, which motivate our model architecture.

3.1. Traffic network

We represent a traffic network as a graph $G = (V, E, A)$, where $V = \{v_1, \dots, v_N\}$ denotes the set of nodes, each corresponding to a specific region, and N is the total number of regions. The edge set E defines the connections between adjacent regions, while $A \in \mathbb{R}^{N \times N}$ denotes the

Table 1

Notations and symbol descriptions.

Symbol	Description
N	Total number of regions or grid cells
M	Length of the input (historical) time segment
P	Length of the prediction (forecast) time segment
G	Traffic network graph structure
ϵ	External factors (e.g., weather, events)
X_w, X_d, X_r	Historical OD demand data: weekly-period, daily-period, and recent-period data
X_M	Historical demand data concatenated from three cycle types (X_w, X_d, X_r)
X	Abbreviation of X_M
D	Feature/channel dimension in the model
C	Number of graph convolution channels
h	Number of attention heads in Transformer
h'	Number of temporal convolution heads
E_{week}	Day-of-week temporal attribute
E_{hour}	Hour-of-day temporal attribute

adjacency matrix of G . Specifically, the adjacency matrix is defined as:

$$A(i, j) = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

here, two regions are considered connected if there is a physical link between them in the traffic network.

3.2. OD demand prediction

In this study, we aim to forecast future OD demands based on historical OD demand observations. The OD demand matrix captures the daily demand of urban populations and exhibits clear periodic characteristics. Specifically, daily variations in travel demand across regions often follow consistent patterns, particularly on weekdays. For example, demand on Mondays and Tuesdays tends to be similar. Moreover, demand distributions on recurring days (e.g., every Monday) demonstrate notable regularity. In addition to periodicity, recent short-term demand trends also provide valuable signals for real-time forecasting.

To exploit these properties, we segment historical OD flow data into three distinct temporal components: weekly-period data $X_w \in \mathbb{R}^{N \times N \times t_w}$, daily-period data $X_d \in \mathbb{R}^{N \times N \times t_d}$, and recent-period data $X_r \in \mathbb{R}^{N \times N \times t_r}$, where t_w , t_d , and t_r denote the lengths of their respective time spans. These temporal segments are commonly adopted in traffic forecasting tasks (Zhang et al., 2017).

Inspired by the connection sampling strategy introduced by Guo et al. (2020), we concatenate these segments to construct a unified temporal input:

$$X_M = \text{concatenate}(X_w, X_d, X_r) \in \mathbb{R}^{N \times N \times M},$$

where $M = t_w + t_d + t_r$ is the total number of temporal slices across all segments. This step is illustrated in Fig. 2 as part of the input module.

Our goal is to predict the OD demand for the next P time steps by leveraging the constructed temporal input X_M , the traffic network G , and external contextual information ϵ . The prediction task can be formally defined as:

$$\hat{X}_{t+1, \dots, t+P} \leftarrow f(X_M, \epsilon, G), \quad (2)$$

where $f(\cdot)$ denotes the predictive function to be learned. For simplicity, we use $X \in \mathbb{R}^{N \times N \times M}$ to denote X_M in the following sections.

4. Proposed method

Fig. 2 presents the architecture of the proposed DSTCN, which is designed to address the complex and dynamic nature of OD demand prediction through three complementary modules. The Glstm2D module simultaneously models spatial and temporal dependencies from both origin and destination perspectives, enhancing the capture of cross-area

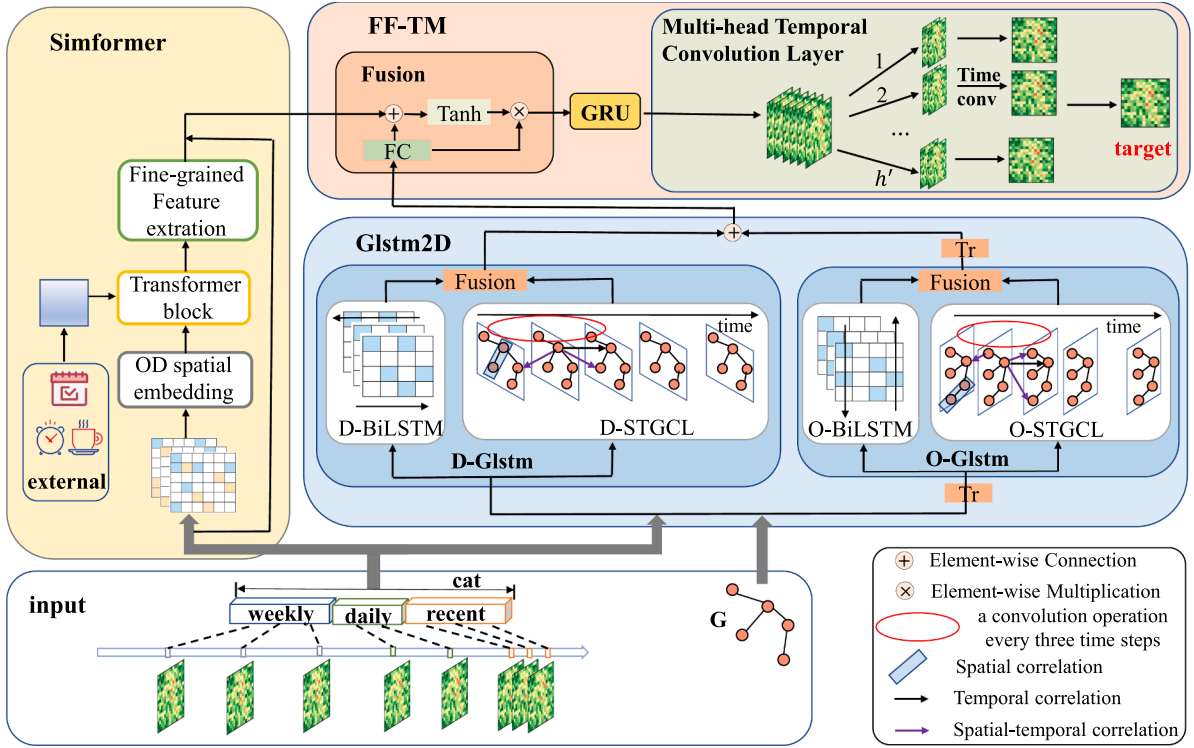


Fig. 2. Overall architecture of the proposed DSTCN framework. The model comprises three key modules: (1) Glstm2D, which models bidirectional demand change trends along both origin and destination axes; (2) Simformer, a Transformer-based spatial similarity module that dynamically captures inter-regional relationships and integrates external factors; and (3) FF-TM, which fuses multi-source features and performs temporal modeling to enable robust and accurate OD demand prediction.

flow patterns. The Simformer module employs Transformer-based attention to dynamically learn inter-region similarities and flexibly incorporate external contextual factors. Finally, the FF-TM module fuses the multi-source features using GRU and temporal convolution, enabling robust and accurate OD demand forecasting.

4.1. Modeling demand change trends in two directions (glstm2d)

Effective OD demand prediction requires capturing the strong, bidirectional correlations between origin and destination regions, a challenge often overlooked by unidirectional or independent modeling approaches. To address this, we introduce the Glstm2D module, which is designed for parallel learning of demand variation trends along both axes. As illustrated in the blue section of Fig. 2, Glstm2D comprises two components: D-Glstm (destination-focused) and O-Glstm (origin-focused). Each employs bidirectional long short-term memory (BiLSTM) networks in combination with spatio-temporal graph learning (STGL), enabling the model to jointly capture reciprocal inter-regional dependencies and temporal evolution. This structure provides an effective mechanism for modeling complex urban mobility patterns. The following section briefly reviews the BiLSTM fundamentals underlying this module.

4.1.1. Basic BiLSTM

OD demand exhibits pronounced tidal asymmetry and phase lags: the dominant commuting direction often reverses between morning and evening peaks, and aggregation/dispersion effects unfold with distinct delays. Relying on a single causal direction can bias representations toward the currently dominant flow. Therefore, we employ a BiLSTM to encode each time step with both its preceding and succeeding context within the observed history window, effectively capturing short- and long-term dependencies.

The LSTM unit comprises a forget gate f_v , input gate i_v , current cell state c_v , output gate o_v , and hidden state h_v , computed as follows:

$$f_v = \sigma([h_{v-1}, x_v]W_f + b_f), \quad (3)$$

$$i_v = \sigma([h_{v-1}, x_v]W_i + b_i), \quad (4)$$

$$c_v = f_v \odot c_{v-1} + i_v \odot \tanh([h_{v-1}, x_v]W_c + b_c), \quad (5)$$

$$o_v = \sigma([h_{v-1}, x_v]W_o + b_o), \quad (6)$$

$$h_v = o_v \odot \tanh(c_v), \quad (7)$$

where $\odot$ denotes the Hadamard product, σ is the sigmoid activation function, and W_*, b_* are learnable parameters.

Let $\bar{X}_f$ denote the input sequence and $\bar{X}_b$ its reversed order. The forward and backward LSTM process these sequences as:

$$\bar{h}_f = LSTM_f(\bar{X}_f), \quad (8)$$

$$\bar{h}_b = LSTM_b(\bar{X}_b), \quad (9)$$

and their outputs are concatenated to form the BiLSTM representation:

$$H_i = \text{concatenate}(\bar{h}_f, \bar{h}_b). \quad (10)$$

Here, $\bar{h}_f \in \mathbb{R}^D$ and $\bar{h}_b \in \mathbb{R}^D$ are the forward and backward hidden representations, and $H_i \in \mathbb{R}^{2D}$ is the resulting feature passed to subsequent modules. We adopt concatenation fusion, which is both expressive, preserving the full subspace spanned by the forward and backward states for downstream mixing, and stable, yielding an effective accuracy–efficiency trade-off for robust trend modeling within Glstm2D.

4.1.2. D-Glstm

The D-Glstm module is designed to capture demand change tendencies along the destination axis by jointly modeling sequential and spatial dependencies. This is achieved by integrating a destination-directed BiLSTM (D-BiLSTM) with a spatio-temporal graph learning component (D-STGL).

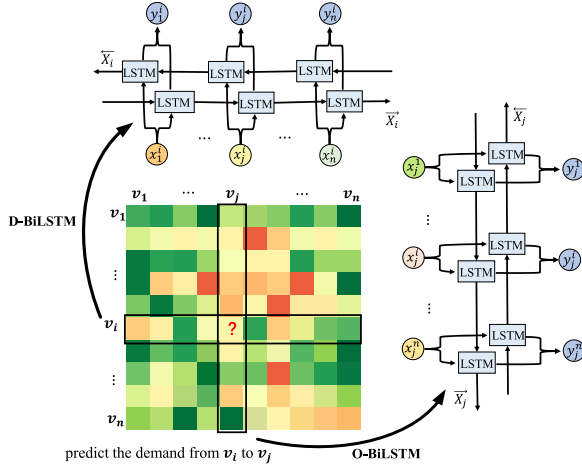


Fig. 3. Illustration of the BiLSTM modeling in both destination and origin directions. The central OD matrix highlights the prediction from v_i to v_j , where D-BiLSTM (top) learns demand trends across destinations for each origin, and O-BiLSTM (right) learns trends across origins for each destination. This bidirectional structure enables comprehensive modeling of OD demand changes.

D-BiLSTM module. The D-BiLSTM learns demand variation trends from each area to all possible destinations, as illustrated in Fig. 3. For the input tensor $X \in \mathbb{R}^{N \times N \times M}$, we denote $X_{iD} \in \mathbb{R}^{1 \times N \times M}$ as the historical demand from area i to all destination areas across M time steps, and the set $\{X_{iD}\}_{i=1}^N$ contains the OD demand sequences for all origin regions. Each X_{iD} is processed by a BiLSTM in the destination direction:

$$H_{iD} = \text{BiLSTM}(X_{iD}), \quad (11)$$

where $H_{iD} \in \mathbb{R}^{1 \times N \times 2D}$ represents the extracted bidirectional features for origin i . Aggregating over all origins yields $H_D \in \mathbb{R}^{N \times N \times 2D}$, which encodes the latent demand change patterns from the destination perspective.

D-STGL module. In practical urban settings—such as adjacent business areas during working days—neighboring regions often experience similar demand fluctuations due to shared functions and synchronized activity patterns. Motivated by this observation, we introduce the D-STGL module, which aggregates demand features from each area and its spatial neighbors across consecutive time steps through spatial graph convolution.

The process begins by constructing a normalized adjacency matrix $\tilde{A} \in \mathbb{R}^{N \times N}$ based on observed OD flows:

$$\tilde{A}(i, j) = \frac{A(i, j)}{\sum_{k=1}^N A(i, k)}, \quad (12)$$

where $\tilde{A}(i, j)$ quantifies the relative strength of travel from area i to j .

Capturing both spatial and short-term temporal dependencies is achieved by applying graph convolution over sliding windows of three consecutive time steps. Specifically, let $X_c = \{X_c^1, \dots, X_c^N\} \in \mathbb{R}^{N \times N \times 3}$ denote the OD demand features from each area i to all others at times $t-1$, t , and $t+1$, with $X_c^t \in \mathbb{R}^{1 \times N \times 3}$. The graph convolution for area i is defined as:

$$HG_c^t = \Theta_{dI} X_c^t + \sum_{j \in Q_{\tilde{A}}(i)} \tilde{A}(i, j) \odot \Theta_{dA} X_c^j, \quad (13)$$

where $\odot$ denotes the Hadamard product, Θ_{dI} and Θ_{dA} are learnable parameters, and $Q_{\tilde{A}}(i)$ represents the neighbor set of i .

By aggregating the output across all areas, we obtain $HG_c = \{HG_c^1, \dots, HG_c^N\} \in \mathbb{R}^{N \times C \times 3}$. Iterating this operation over all $M-2$ sliding windows yields the complete spatio-temporal feature tensor:

$$HG = \{HG_1, \dots, HG_{M-2}\} \in \mathbb{R}^{N \times C \times 3 \times (M-2)}. \quad (14)$$

Finally, to comprehensively exploit both sequential and spatial context, the neighborhood-aware features HG are fused with the D-BiLSTM

output H_D as follows:

$$X_{DGlstm} = H_D \cdot \tanh(FC(H_D \oplus FC(HG))), \quad (15)$$

where $\oplus$ denotes feature concatenation and FC is a fully connected layer. The resulting $X_{DGlstm} \in \mathbb{R}^{N \times N \times 2D}$ encodes refined OD demand tendencies in the destination direction.

4.1.3. O-Glstm and combination strategy

To capture the full spectrum of OD demand evolution, we complement the D-Glstm module with O-Glstm, which is specifically designed to model demand variation from all origins to each destination. Although the architectural design of O-Glstm mirrors that of D-Glstm, both comprising a BiLSTM and a spatio-temporal graph learning component, their modeling orientations differ. In O-Glstm, we transpose the original OD feature tensor so that each slice $X_c^{iT} \in \mathbb{R}^{N \times 1 \times 3}$ now represents the demand from all origins to a single destination i over consecutive time steps.

Passing the transposed input X_c^T through the O-BiLSTM and O-STGL modules yields the O-Glstm output $X_{OGlstm} \in \mathbb{R}^{N \times N \times 2D}$, where the first dimension indexes destinations and the second indexes origins. To ensure consistency with the (origin, destination) semantics of X_{DGlstm} , we transpose the first two dimensions of X_{OGlstm} to obtain $X'_{OGlstm} \in \mathbb{R}^{N \times N \times 2D}$, so that the first dimension corresponds to origins and the second to destinations.

Finally, to form a comprehensive representation that captures demand change characteristics from both perspectives, we concatenate X_{DGlstm} and X'_{OGlstm} along the feature dimension and apply a fully connected layer:

$$X_{Glstm} = FC(X_{DGlstm} \oplus X'_{OGlstm}), \quad (16)$$

where $\oplus$ denotes feature concatenation and FC is a fully connected transformation. The resulting $X_{Glstm} \in \mathbb{R}^{N \times N \times D}$ provides a unified embedding that encodes bidirectional OD demand change features, serving as a robust foundation for downstream prediction.

4.2. Transformer-based spatial similarity module (simformer)

OD grid cells within the matrix often exhibit dynamic and context-dependent demand similarities, as analyzed in Section 5.10. Effectively leveraging these similarities can enhance the prediction accuracy for regions exhibiting analogous demand patterns. To this end, we propose a Transformer-based spatial similarity (Simformer) module, which is specifically designed to capture and model the shared features among grid cells in the OD matrix.

As depicted in Fig. 4, Simformer operates through four key stages: OD spatial embedding, extraction of external features, Transformer-based context modeling, and fine-grained feature extraction. This architecture enables the network to exploit both explicit and implicit spatial similarities for improved OD demand prediction.

4.2.1. OD spatial embedding

The OD demand matrix for large-scale urban datasets, such as the 69×69 grid in Manhattan, typically contains a large number of grid regions, resulting in significant computational complexity. To address this, we aggregate the feature information of adjacent grid cells via a convolutional operation, inspired by the design in Peng et al. (2021). Specifically, we group each 4×4 block of OD grid cells into a single cell, using the original tensor $X \in \mathbb{R}^{N \times N \times M}$ as input to Simformer.

This process produces an embedded tensor $X_{conv} \in \mathbb{R}^{N_{se} \times N_{se} \times D}$, where $N_{se} = N/4$. Thus, the spatial resolution of the OD matrix is reduced while the number of feature channels is increased. To facilitate interaction modeling among all OD regions, X_{conv} is further flattened into $X_{flatten} \in \mathbb{R}^{N_g \times D}$, where $N_g = N_{se} \times N_{se}$. This representation enables efficient global context modeling in subsequent Transformer layers.

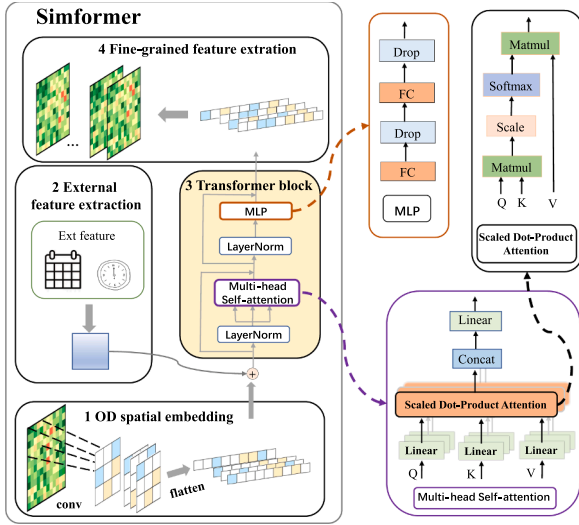


Fig. 4. Simformer module.

Although the OD matrix is flattened to $X_{Flatten} \in \mathbb{R}^{N_g \times D}$ for efficiency, each token remains index informed before entering the Transformer. As detailed in Section 4.1, the preceding Glstm2D provides dual-stream encodings along rows (origins) and columns (destinations), yielding origin-wise and destination-wise summaries u_i and v_j . We form the cell representation as

$$x_{ij} = \psi(u_i, v_j, e_{ij}),$$

where e_{ij} denotes cell-specific exogenous features. Thus, x_{ij} already carries the identities of row i and column j , the subsequent self-attention operates on the index content.

4.2.2. External feature extraction

Incorporating temporal attributes into grid cell representations can significantly enhance the model's ability to capture temporal patterns in OD demand. For each grid cell g_j^i at time interval t , we encode external temporal features, specifically, the day of the week and the hour of the day, using one-hot encoding (Ye et al., 2021). The day-of-week attribute is mapped to a 7-dimensional vector $E_{week} \in \mathbb{R}^7$, while the hour-of-day attribute is mapped to a 24-dimensional vector $E_{hour} \in \mathbb{R}^{24}$.

To align the dimension of these external features with the demand feature space, both E_{week} and E_{hour} are projected to D -dimensional embeddings via fully connected layers, resulting in $Ef_{week} \in \mathbb{R}^D$ and $Ef_{hour} \in \mathbb{R}^D$. These embeddings are then broadcast to match the number of OD grid cells, yielding $Ef'_{week} \in \mathbb{R}^{N_g \times D}$ and $Ef'_{hour} \in \mathbb{R}^{N_g \times D}$.

The final input to the Transformer, $X_{next} \in \mathbb{R}^{N_g \times D}$, is constructed by combining the flattened OD demand features $X_{Flatten}$ with the embedded temporal features as follows:

$$X_{next} = X_{Flatten} + \text{ReLU}(Ef'_{week} + Ef'_{hour}), \quad (17)$$

where all terms share the same hidden dimension D . This enriched representation enables the model to leverage both demand and temporal context in the subsequent Transformer block for learning dynamic correlations between OD grids.

4.2.3. Transformer block

The Transformer block processes the enriched grid cell representations X_{next} to model their dynamic inter-relationships. It employs a standard multi-head self-attention mechanism, followed by an MLP block, with LayerNorm and residual connections applied throughout to ensure stable training.

For the attention computation, the input is linearly projected to generate queries (Q), keys (K), and values (V) for h parallel heads,

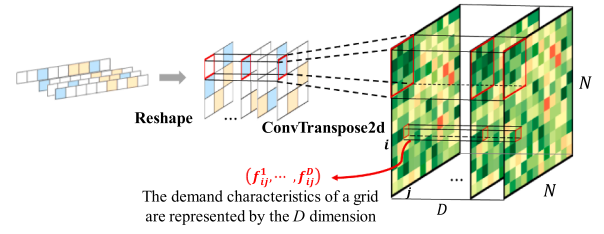


Fig. 5. Fine-grained feature extraction process.

each with its own latent space: $K \in \mathbb{R}^{h \times N_g \times D/h}$, $Q \in \mathbb{R}^{h \times N_g \times D/h}$, and $V \in \mathbb{R}^{h \times N_g \times D/h}$. These are computed as:

$$K = (K_1, \dots, K_h) = f_{key}(X_{next}), \quad (18)$$

$$Q = (Q_1, \dots, Q_h) = f_{query}(X_{next}), \quad (19)$$

$$V = (V_1, \dots, V_h) = f_{value}(X_{next}), \quad (20)$$

where f_{key} , f_{query} , and f_{value} are learned linear projections.

For each head, the attention weights are calculated using scaled dot-product attention:

$$\text{Attn}(Q_i, K_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right), \quad (21)$$

where $d_k = D/h$ is the dimension of each head. The resulting attention matrix $\text{Attn}(Q_i, K_i) \in \mathbb{R}^{N_g \times N_g}$ quantifies the pairwise influence between OD grids. And the softmax function normalizes these influence scores across all grid cells for each query, converting them into a set of positive weights that sum to one. This ensures that the attention mechanism learns a valid probability distribution over all other cells, representing their relative importance.

The attended value representations are computed as:

$$\text{head}_i = \text{Attn}(Q_i, K_i) V_i \in \mathbb{R}^{N_g \times D/h}, \quad (22)$$

and all heads are concatenated to form the output of the multi-head attention layer:

$$X_{trans} = \text{concat}(\text{head}_1, \dots, \text{head}_h) \in \mathbb{R}^{N_g \times D}. \quad (23)$$

This output is subsequently processed by an MLP block, consisting of an up-projection and a down-projection fully connected layer, to yield $X_{mlp} \in \mathbb{R}^{N_g \times D}$ Fig. 5

4.2.4. Fine-grained feature extraction

The goal of this step is to recover high-resolution, fine-grained features for each OD grid cell from the compressed representation generated by the Transformer block. Given $X_{mlp} \in \mathbb{R}^{N_g \times D}$, we first reshape it to $X_{shape} \in \mathbb{R}^{N_{se} \times N_{se} \times D}$, where $N_{se} = N/4$ matches the reduced spatial size after the convolutional embedding.

To recover a high resolution feature map from the spatially coarse but semantically rich output of the Transformer, we employ a multi stage coarse-to-fine restoration process. The primary step applies a transposed convolution, which serves as a learnable upsampling operator to spatially expand the feature map from the reduced $N_{se} \times N_{se}$ resolution back to the original $N \times N$ scale, restoring the spatial grid structure while preserving the high level correlation patterns learned by the self attention mechanism. The resulting tensor is $X_{deconv} \in \mathbb{R}^{N \times N \times D}$.

To further preserve information from the original OD demand matrix, we incorporate a residual connection between X_{deconv} and a projected version of the original feature X :

$$X_{simformer} = X_{deconv} + FC(X), \quad (24)$$

where FC denotes a fully connected layer mapping the original input X to the same feature dimension. The resulting $X_{simformer} \in \mathbb{R}^{N \times N \times D}$ fuses both the original and the enhanced fine-grained features, providing a rich and stable representation for subsequent prediction tasks.

4.3. Feature fusion and time modeling (FF-TM)

Building on the extracted spatial and grid-level features, the Feature Fusion and Time Modeling (FF-TM) module integrates these representations and models their temporal evolution. FF-TM comprises three key components: spatio-temporal feature fusion, GRU-based temporal modeling, and a multi-head temporal convolution layer.

4.3.1. Spatio-temporal feature fusion

Following the extraction of area demand change features by the Glsim2D module and the OD grid features by the Simformer module, we proceed to fuse these heterogeneous representations. Inspired by the concatenation-based fusion approach of Liu et al. (2020), we concatenate the outputs X_{Glsim} and $X_{simformer}$, and map the concatenated features to a comprehensive hidden representation H using a fully connected layer:

$$H = FC(X_{Glsim} \oplus X_{simformer}), \quad (25)$$

where $\oplus$ denotes feature concatenation. The resulting H effectively integrates both area demand change information and fine-grained OD grid cell characteristics.

To further enhance feature expressiveness, we apply a Tanh activation to H and fuse it multiplicatively with X_{Glsim} , followed by a fully connected layer to obtain the final fused feature $U \in \mathbb{R}^{N \times N \times M}$:

$$U = FC(X_{Glsim} \cdot \tanh(H)), \quad (26)$$

where $\cdot$ denotes element-wise multiplication. In Section 5.6, we empirically demonstrate the effectiveness of this fusion strategy. The fused feature U serves as the input for the subsequent GRU-based temporal modeling module.

4.3.2. GRU time modeling

To model the temporal dynamics of OD demand, we employ Gated Recurrent Units (GRUs) to capture the time-varying relationships within the fused feature sequence $U = [U_1, \dots, U_M]$. Compared to LSTM, GRU offers a more compact architecture with fewer parameters, thus facilitating faster convergence and efficient sequence modeling.

The GRU cell is computed as follows:

$$r_t = \sigma([H_{t-1}, U_t]W_r + b_r), \quad (27)$$

$$z_t = \sigma([H_{t-1}, U_t]W_u + b_u), \quad (28)$$

$$\tilde{H}_t = \tanh(r_t \odot [H_{t-1}, U_t]W_h + b_h), \quad (29)$$

$$H_t = z_t \odot H_{t-1} + (1 - z_t) \odot \tilde{H}_t, \quad (30)$$

where r_t and z_t denote the reset and update gates, respectively, and $\odot$ is the Hadamard (element-wise) product. $H_t \in \mathbb{R}^{D \times N}$ represents the hidden state at time t , U_t is the fused input at time t , and $\sigma(\cdot)$ denotes the sigmoid activation function. W_r, W_u, W_h and b_r, b_u, b_h are trainable parameters.

By sequentially applying the GRU cell across all time steps, we obtain the temporal representation $X_{gru} \in \mathbb{R}^{D \times N \times M}$, which encodes the time-dependent evolution of OD demand features for subsequent prediction layers.

4.3.3. Multi-head temporal convolution layer

Finally, we introduce a multi-head temporal convolution layer to effectively aggregate temporal information and update the spatio-temporal representations of each OD pair. This layer is designed to merge information from adjacent time slices, thus capturing diverse temporal dependencies within the data.

To model heterogeneous temporal patterns across different intervals, we partition the M temporal segments in X_{gru} into h' groups, with each group comprising M_i consecutive time steps such that $M = M_1 + M_2 + \dots + M_{h'}$. Let $X_{gru}^i \in \mathbb{R}^{D \times N \times M_i}$ denote the i th temporal segment, so that

$$X_{gru} = (X_{gru}^1, \dots, X_{gru}^{h'}).$$

For each segment, we apply a temporal convolution operator with kernel size $1 \times t_i$, where $t_i = M_i/P$ and P is the number of prediction time steps:

$$X_{out}^i = \Phi * X_{gru}^i = \text{Conv}_{1 \times t_i}(X_{gru}^i), \quad (31)$$

where Φ represents the learnable convolution kernel parameters, and $*$ denotes the convolution operation.

The outputs from all heads are aggregated by element-wise summation:

$$X_{out} = X_{out}^1 + \dots + X_{out}^{h'}, \quad (32)$$

where $X_{out}^i \in \mathbb{R}^{N \times N \times P}$ and $X_{out} \in \mathbb{R}^{N \times N \times P}$ constitute the final predictions across all P forecasted time intervals.

This multi-head design enables the model to flexibly capture complex and multi-scale temporal dependencies, thereby improving the accuracy and robustness of OD demand forecasting.

4.3.4. Loss function

To optimize the OD demand prediction model, we adopt the mean squared error (MSE) as the training objective. Let $\hat{X}_t(i, j)$ and $Y_t(i, j)$ denote the predicted and ground-truth OD demand values, respectively, from area i to area j at time step t . The loss function is defined as:

$$L(\hat{X}, Y) = \frac{1}{PN^2} \sum_{t=1}^P \sum_{i=1}^N \sum_{j=1}^N (\hat{X}_t(i, j) - Y_t(i, j))^2, \quad (33)$$

where P is the prediction horizon and N is the total number of areas. This loss encourages the model to minimize the average squared difference between predicted and actual OD demands across all time steps and region pairs.

5. Experiments

In this section, we conduct comprehensive experiments on three widely-used benchmark traffic datasets to evaluate the effectiveness of the proposed model. Detailed dataset statistics are presented in Table 2.

5.1. Experimental settings

5.1.1. Datasets

We conduct experiments on three widely adopted real-world benchmarks: NYC-TOD2018, NYC-TOD2019, and HZMetro. For each dataset, we implement rigorous preprocessing steps to ensure data quality and consistency. Records with invalid timestamps or ambiguous origins/destinations were filtered out, and entries with empty stations, such as the unused 54th station in HZMetro, were removed, resulting in 80 valid stations. To mitigate potential biases arising from scale or sparsity, all OD demand values were normalized using Z-score standardization.

Table 2
Dataset statistics and settings.

Property	NYC-TOD2018	NYC-TOD2019	HZMetro
Areas	69	69	80
Timespan	2018/1/1–2018/12/31	2019/1/1–2019/12/31	2019/1/1–2019/1/25
Time interval	1 h	1 h	10 min
w:d:r ratio	2:1:3	2:1:3	2:1:3
Input times-tamp(s)	6	6	6, 18, 36
Output times-tamp(s)	1	1	1, 3, 6
Forecast time length	60 min	60 min	10 min 30 min 60 min
Mean demand	3.31	2.66	1.68
Std. dev. of demand	9.17	7.67	4.46

- NYC-TOD2018 and NYC-TOD2019 Dataset. These datasets are derived from the yellow taxi trip records provided by the New York City Taxi and Limousine Commission¹ (TLC). Manhattan is divided into 69 regions according to the official TLC zoning. NYC-TOD2018 and NYC-TOD2019 respectively cover taxi demand in Manhattan throughout the years 2018 and 2019. To address demand sparsity in short intervals, we aggregate trips using a one-hour time window, which balances temporal resolution and data density for robust OD flow prediction.
- HZMetro Dataset. The HZMetro dataset originates from the Alibaba Tianchi platform² and consists of approximately 58 million transaction records collected from 80 Hangzhou metro stations between January 1, and January 25, 2019. Each record comprises passenger ID, station, access status, timestamp, and more. Since metro services are suspended from 0:00–6:00, we only consider transactions from 6:00–24:00 each day, and aggregate demand in 10-min intervals (i.e., six periods per hour). All OD data were filtered to exclude records with incomplete or invalid station identifiers.

5.1.2. Implementation details

All datasets are partitioned into training, validation, and test sets with a ratio of 6:2:2. The historical OD demand inputs X_w , X_d , and X_r are constructed with a period ratio of 2:1:3, respectively. For NYC-TOD2018 and NYC-TOD2019, we perform long-term prediction of OD demand for the next hour. For HZMetro, we conduct both short-term and long-term forecasting for the next 10, 30, and 60 min. All experiments are repeated 10 times, and the average results are reported for robust comparison.

Key hyperparameters are validated through ablation in Section 5.8. In our implementation, the graph convolution kernel size is set to 3, and the hidden feature dimension D is fixed at 256. The Simformer employs 6 attention heads, while the GRU shares the same feature dimension. The multi-head temporal convolution uses $h' = 3$ heads.

For model optimization, we use the Mean Squared Error (MSE) loss and the Adam optimizer. The initial learning rate is set to $5e - 4$ with an exponential decay rate of 0.92. Models are trained for 100 epochs with a batch size of 16. All models are implemented in PyTorch.

5.1.3. Evaluation metrics

We evaluate model performance using two widely adopted regression metrics: mean absolute error (MAE) and root mean square error (RMSE), defined as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{X}^i - Y^i|, \quad (34)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{X}^i - Y^i)^2}, \quad (35)$$

where n is the total number of test samples, and $\hat{X}^i$ and Y^i denote the predicted and ground-truth values of the i th test sample, respectively. Prior to evaluation, all predictions are rescaled to their original scale to ensure fair comparison with ground-truth data.

5.1.4. Baselines

To comprehensively evaluate the effectiveness of DSTCN, we compare it with strong baseline methods from three categories: heuristic, state-of-the-art OD demand prediction, and advanced urban flow prediction models.

Heuristic method:

HA (Historical Average): A simple baseline that predicts future demand by averaging historical passenger demand values for each OD pair.

OD demand prediction methods:

GEML (Wang et al., 2019): Aggregates both geographic and semantic neighbor information to enhance the extraction of spatial dependencies.

MPGCN (Shi et al., 2020): Models complex spatial dependencies using one dynamic and two static graphs.

HMOD (Zhang et al., 2022): Employs a dynamic graph representation learning framework for advanced OD demand prediction.

CMOD (Han et al., 2022): Introduces a continuous-time, multi-level dynamic graph representation for OD demand forecasting.

DSTCN_R: A simplified variant of our DSTCN model that utilizes only recent demand data as input features.

Urban flow prediction methods:

Given the limited availability of benchmark OD demand prediction models, we compare with several representative urban traffic forecasting approaches known for their strong performance. To ensure a fair comparison, we adapt these models to the OD demand prediction setting, following their original hyperparameters and codebases wherever possible.

GRU: A recurrent neural network with two stacked GRU layers, each with a hidden size of 256. Other hyperparameters are set to match those of DSTCN.

LSTM: Similar to the GRU baseline, but each recurrent layer is replaced by LSTM units.

STGCN (Yu et al., 2018): Combines ChebNet-based spatial convolutions with temporal convolutions to capture spatio-temporal dependencies. Adapted from its official implementation for the OD demand prediction task.

STSGCN (Song et al., 2020): Employs a spatio-temporal synchronous graph convolutional network to extract local correlations. Applied to OD demand prediction as per its official code.

DGCN (Guo et al., 2020): Enhances traditional graph convolution with a latent network for adaptive dynamic graph construction. Adapted for OD prediction following the official implementation.

STTN (Xu et al., 2020): Applies Transformers to model both spatial and temporal correlations in urban demand. Adapted to the OD setting based on the official codebase.

TGCRN (Ma et al., 2024): Combines time-aware graph structure learning with graph convolutional GRUs to capture dynamic trends and periodic patterns in spatially correlated time series for prediction.

5.2. Results on NYC-TOD

We first assess the performances of our model and competitive methods on the NYC-TOD dataset. In all result tables, the overall best result is highlighted in **bold**, while the best baseline result (excluding DSTCN and DSTCN_R) is underlined. Percentage improvements (Δ) are calculated relative to this best baseline.

5.2.1. Results on the city-wide network

Table 3 reports the MAE and RMSE of all compared methods on NYC-TOD2018 and NYC-TOD2019. Our proposed DSTCN achieves the best performance across all metrics and datasets, with improvements being statistically significant ($p < .001$). 1) Methods that explicitly model spatial dependencies, such as STGCN and DGCN, consistently outperform time-only models (e.g., GRU, LSTM), highlighting the importance of spatial correlation in OD demand prediction. 2) While several state-of-the-art OD demand models (e.g., GEML, MPGCN, STTN) incorporate both spatial and temporal cues, they still exhibit performance gaps. Specifically, GEML relies on manually defined spatial dependencies, MPGCN overlooks correlations between regions with similar demand patterns, and STTN does not fully exploit the interplay between spatial and temporal contexts. 3) It is noteworthy that general urban flow models (STGCN, DGCN) surpass dedicated OD prediction methods (MPGCN, CMOD) in most cases. This suggests that modeling multi-scale traffic networks (STGCN) or learning latent spatio-temporal features (DGCN) is beneficial for OD prediction, likely due to the transferable nature of traffic pat-

¹ <https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

² <https://tianchi.aliyun.com/>

Table 3

Performance Comparison on NYC-TOD2018 and NYC-TOD2019. Improvements of DSTCN over the best baseline are shown in the last row.

Model	NYC-TOD 2018		2019	
	MAE	RMSE	MAE	RMSE
HA	1.973	4.944	1.718	4.419
GRU	1.586	3.613	1.365	3.160
LSTM	1.569	3.568	1.350	3.147
STTN	1.529	3.512	1.317	3.132
STSGCN	1.535	3.473	1.315	3.087
DGCN	1.534	3.441	1.327	3.033
STGCN	<u>1.515</u>	3.483	1.317	3.110
TGCRN	1.525	3.722	1.336	3.132
GEML	1.787	4.071	1.544	3.592
HMOD	1.577	3.655	1.387	3.314
MPGCN	1.549	3.799	1.320	3.362
CMOD	1.542	3.785	1.318	3.411
DSTCN	1.469	3.284	1.273	2.909
DSTCN_R	1.479	3.330	1.277	2.945
Δ	3.04 %	4.56 %	3.19 %	4.09 %

Table 4

Performance on high-demand areas in NYC-TOD2018 and NYC-TOD2019 (Top 20 OD stations).

Model	NYC-TOD2018		NYC-TOD2019	
	MAE	RMSE	MAE	RMSE
HA	7.156	12.050	6.355	10.850
GRU	5.394	8.671	4.671	7.545
LSTM	5.347	8.515	4.632	7.528
STTN	5.277	8.437	4.604	7.515
STSGCN	5.260	8.320	4.561	7.385
DGCN	5.198	<u>8.225</u>	<u>4.530</u>	<u>7.236</u>
STGCN	<u>5.184</u>	8.340	4.554	7.446
TGCRN	5.647	8.498	4.704	7.481
GEML	6.351	10.251	5.547	9.068
HMOD	5.459	8.890	4.816	8.068
MPGCN	5.599	9.313	4.915	8.290
CMOD	5.575	9.618	4.908	8.814
DSTCN	4.952	7.752	4.326	6.879
DSTCN_R	4.987	7.934	4.358	7.024
Δ	4.48 %	5.75 %	4.50 %	4.93 %

terns. 4) DSTCN yields the lowest MAE and RMSE on both datasets. This superiority is attributed to our unified modeling of bidirectional spatial and temporal dependencies, which more faithfully reflects the complex dynamics of urban mobility. Concretely, DSTCN achieves average improvements of 3.04 %, 4.56 %, 3.19 %, and 4.09 % in MAE and RMSE on NYC-TOD2018 and NYC-TOD2019, respectively, over the best baseline. Furthermore, even the DSTCN_R variant, which utilizes only recent data, outperforms most baselines, further corroborating the robustness and effectiveness of our design for real-world OD demand forecasting.

5.2.2. Results on the high-demand areas

In practical urban transportation systems, regions with high traffic demand are of particular importance, as they serve larger populations and exhibit greater temporal variability. Accurate OD demand prediction in these areas is essential for effective resource allocation and congestion management. However, the inherent volatility of these regions presents additional forecasting challenges.

To rigorously evaluate model robustness under such conditions, we identify the top 20 stations with the highest cumulative departure demand in the NYC-TOD2018 and NYC-TOD2019 datasets and extract the corresponding OD submatrices. We then benchmark the predictive performance of all competing methods on these high-demand OD pairs.

As summarized in Table 4, DSTCN outperforms all baselines in both MAE and RMSE. Specifically, DSTCN achieves relative improvements of **4.48 %** and **5.75 %** (MAE and RMSE) for NYC-TOD2018, and **4.50 %** and **4.93 %** for NYC-TOD2019, respectively, over the best-performing

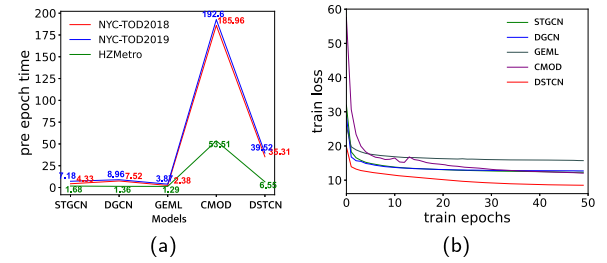


Fig. 6. Efficiency comparison. (a) Average training time (sec/epoch). (b) Convergence curves on NYC-TOD2018.

state-of-the-art methods. These results demonstrate the superior capability of DSTCN to capture complex spatio-temporal dependencies and adapt to highly dynamic demand environments, thereby ensuring robust prediction performance where it matters most in urban mobility management.

5.2.3. Results in different time scenarios

To thoroughly evaluate model robustness, we compare the prediction accuracy of all methods across three representative periods: morning rush (M-rush), off-peak (Non-rush), and evening rush (E-rush). Tables 5 and 6 summarize the MAE and RMSE results for the NYC-TOD2018 and NYC-TOD2019 datasets, separately for weekdays and weekends.

Notably, HA and GEML methods are less stable, and MPGCN struggles to handle rapid demand changes. State-of-the-art spatio-temporal models such as STGCN and DGCN show strong performance during periods of fluctuating demand. DSTCN achieves the best or near-best performance across all periods and both datasets, demonstrating its effectiveness and stability in both peak and non-peak scenarios. DSTCN is particularly robust in capturing both sudden surges and steady trends in OD demand.

5.3. Results on HZMetro

To further evaluate the transferability of DSTCN, we tested both long- and short-term forecasting on the HZMetro dataset. As shown in Table 7, DSTCN consistently achieves the lowest MAE and RMSE across all forecasting horizons, outperforming all competing baselines. Notably, even DSTCN_R surpasses all other baseline models, further demonstrating the robustness of our approach for metro OD prediction in both short- and long-term scenarios.

Overall, through extensive experiments on three datasets, DSTCN consistently delivers the best performance in various settings. These results highlight the model's versatility and strong generalization capability for both road-based and metro OD demand prediction by effectively capturing spatio-temporal correlations.

5.4. Efficiency comparison

We evaluate the computational efficiency of DSTCN against representative baselines, with results summarized in Table 8 and Fig. 6. The analysis shows that DSTCN intentionally trades a higher computational budget for superior accuracy. It has a larger parameter count compared to lightweight baselines like STGCN, which contributes to its moderate training time (Fig. 6 (a)).

Nevertheless, the model remains highly practical. Its inference latency is 1.204 ms/sample, which is suitable for online deployment. A further advantage is training stability, as DSTCN achieves the fastest and most stable convergence among all tested models (Fig. 6(b)).

Overall, DSTCN achieves a favorable accuracy–efficiency profile, maintaining practical inference speed and fast, stable convergence despite its larger parameter count.

Table 5

Performance comparison on NYC-TOD2018 by Time Periods (MAE/RMSE). Periods: M-rush (7:00-9:00), Non-rush (14:00-16:00), E-rush (17:00-19:00) for weekdays and weekends.

Model	Weekdays						Weekends					
	M-rush		Non-rush		E-rush		M-rush		Non-rush		E-rush	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	2.214	5.821	2.295	5.574	2.620	6.229	1.256	3.193	2.605	5.811	2.609	5.739
GRU	1.833	4.419	1.820	4.152	1.983	4.467	1.143	2.597	1.998	4.194	2.041	4.354
LSTM	1.808	4.311	1.791	3.999	1.961	4.340	1.125	2.589	1.984	4.135	2.012	4.247
STTN	1.713	4.163	1.773	3.895	1.965	4.282	1.052	2.511	2.008	4.230	2.018	4.206
STSGCN	1.681	3.851	1.762	3.872	1.984	4.356	1.079	2.502	2.011	4.164	2.037	4.219
TGCRN	1.793	4.629	1.805	4.033	2.073	4.607	1.102	2.601	1.994	4.323	2.042	4.270
DGCN	1.795	4.028	1.766	3.894	1.966	4.228	1.042	2.354	1.968	4.125	1.985	4.084
STGCN	1.669	3.815	1.754	3.882	1.956	4.319	1.048	2.436	2.006	4.299	2.010	4.220
GEML	2.078	4.977	1.923	4.304	2.232	4.960	1.380	3.034	2.252	5.182	2.213	4.627
HMOD	1.786	4.243	1.844	4.239	2.061	4.721	1.097	2.557	2.045	4.434	2.059	4.356
MPGCN	1.929	5.044	1.796	4.353	2.003	4.823	1.082	2.597	1.982	4.158	1.972	4.362
CMOD	1.776	4.578	1.831	4.499	2.081	4.892	1.062	2.454	2.023	4.753	2.035	4.462
DSTCN	1.579	3.494	1.725	3.756	1.933	4.095	0.989	2.231	1.909	3.983	1.945	3.975
DSTCN_R	1.636	3.712	1.711	3.802	1.927	4.103	1.029	2.259	1.933	4.048	1.985	4.014
△	5.39 %	8.41 %	2.45 %	3.00 %	1.48 %	3.15 %	5.09 %	5.23 %	3.00 %	3.44 %	1.37 %	2.67 %

Table 6

NYC-TOD2019. Comparison of Different Periods. We report MAE and RMSE for all test data. Periods: M-rush (7:00-9:00), Non-rush (14:00-16:00), E-rush (17:00-19:00).

Model	weekdays						weekends					
	M-rush		Non-rush		E-rush		M-rush		Non-rush		E-rush	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.791	4.942	2.137	5.248	2.411	5.717	1.298	3.520	2.087	4.874	2.173	4.994
GRU	1.431	3.639	1.635	3.639	1.813	4.013	1.173	2.977	1.622	3.569	1.687	3.667
LSTM	1.425	3.585	1.623	3.655	1.807	4.026	1.173	2.950	1.613	3.592	1.674	3.657
STTN	1.365	3.577	1.617	3.656	1.799	3.991	1.088	2.899	1.613	3.648	1.671	3.669
STSGCN	1.334	3.335	1.596	3.544	1.787	3.932	1.109	2.842	1.605	3.561	1.678	3.640
DGCN	1.343	3.026	1.619	3.550	1.792	3.844	1.104	2.668	1.627	3.566	1.667	3.596
STGCN	1.321	3.124	1.633	3.712	1.804	4.037	1.086	2.654	1.635	3.711	1.678	3.697
TGCRN	1.472	3.763	1.694	3.683	1.894	4.018	1.277	2.832	1.638	3.627	1.697	3.633
GEML	1.665	4.114	1.759	4.085	1.939	4.272	1.425	3.365	1.794	4.255	1.834	4.042
HMOD	1.426	3.509	1.706	3.973	1.910	4.422	1.157	2.884	1.688	3.897	1.738	3.899
MPGCN	1.462	4.107	1.616	3.943	1.818	4.346	1.136	3.258	1.578	3.730	1.683	3.754
CMOD	1.339	3.737	1.653	4.324	1.869	4.635	1.077	2.938	1.621	4.013	1.712	3.976
DSTCN	1.242	2.845	1.584	3.420	1.760	3.716	1.030	2.478	1.571	3.414	1.648	3.473
DSTCN_R	1.282	3.110	1.568	3.431	1.754	3.736	1.057	2.597	1.568	3.454	1.647	3.513
△	5.98 %	5.98 %	1.75 %	3.50 %	1.85 %	3.33 %	4.36 %	6.63 %	0.63 %	4.13 %	1.20 %	3.42 %

Table 7

Performance Comparison on HZMetro.

Model	HZMetro					
	10min		30min		60min	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	1.168	2.509	1.169	2.510	1.169	2.509
GRU	1.145	2.280	1.079	2.141	1.179	2.369
LSTM	1.119	2.253	1.055	2.115	1.147	2.313
STTN	1.061	2.204	1.073	2.212	1.069	2.210
STSGCN	1.045	2.154	1.053	2.177	1.072	2.243
DGCN	1.046	2.013	1.058	2.009	1.060	2.038
STGCN	1.035	2.056	1.046	2.117	1.079	2.304
TGCRN	1.075	2.122	1.067	2.198	1.092	2.299
GEML	1.191	2.482	1.172	2.288	1.221	2.375
HMOD	1.065	2.204	1.083	2.185	1.113	2.249
MPGCN	1.030	2.048	1.041	2.119	1.082	2.293
CMOD	1.075	2.133	1.070	2.157	1.109	2.238
DSTCN	0.931	1.827	0.924	1.832	0.935	1.827
DSTCN_R	0.950	1.838	0.962	1.863	0.990	1.933
△	9.61 %	9.24 %	11.24 %	8.81 %	11.79 %	10.35 %

Table 8

Efficiency comparison on the NYC-TOD2018 dataset under identical settings.

Model	Parameters (M)	Avg. Inference (ms/sample)
STGCN	0.18	0.217
DGCN	0.77	0.203
GEML	2.50	0.119
CMOD	0.30	2.388
DSTCN (Ours)	14.42	1.204

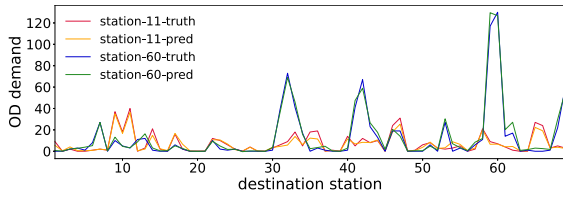
Fig. 7(a) compares predicted and actual demand changes for two areas, illustrating that D-BiLSTM closely tracks real demand trends. As the LSTM cells process demand sequences for all destination areas, the resulting features comprehensively represent origin-to-destination temporal evolution. Importantly, since destination indices are not necessarily geographically adjacent, reordering these indices alters the input trend patterns. Nevertheless, as shown in Fig. 7(b), even after destination order is shuffled, D-BiLSTM can still reconstruct demand trends matching the ground truth when outputs are remapped to the original order. This demonstrates BiLSTM's robustness in learning demand correlations independent of region ordering.

5.6. Ablation study

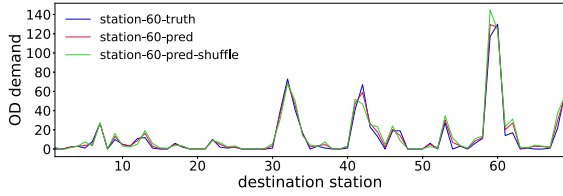
We conduct ablation studies on the NYC-TOD2018 and NYC-TOD2019 datasets, with results summarized in Table 9. The ablations

5.5. The effectiveness of BiLSTM in capturing demand correlation

We examine how BiLSTM captures OD demand trend correlations, using the D-BiLSTM module as an example. Owing to its bidirectional architecture, BiLSTM effectively models global and local variation patterns, capturing commonalities and distinct dynamics across regions.



(a) Demand trends for departures from areas 11 and 60.



(b) Impact of destination order on D-BiLSTM predictions, with reordered destination sequences.

Fig. 7. Comparison of real and D-BiLSTM-predicted demand values for December 31, 16:00–17:00 on the NYC-TOD2018 dataset. (a) Demand trends for departures from areas 11 and 60. (b) Impact of destination order on D-BiLSTM predictions, with reordered destination sequences.

include removing specific components: “w/o D-BiLSTM” and “w/o D-STGL” remove the D-BiLSTM or D-STGL from the D-Glstm2D module, while “w/o O-BiLSTM” and “w/o O-STGL” do so for O-Glstm2D. “w/o Glstm2D” eliminates both D-Glstm and O-Glstm modules. “w/o External” omits external factors. “w/o Simformer,” “w/o GRU,” and “w/o Mul-TCL” refer to removing the Simformer module, the GRU module, and the multi-head temporal convolution layer, respectively.

Table 9 demonstrates that: 1) All subcomponents within D-Glstm2D and O-Glstm2D (D-BiLSTM, D-STGL, O-BiLSTM, O-STGL) improve prediction, but the integrated Glstm2D achieves the best performance, confirming the benefit of bidirectional modeling. 2) External factors, the Simformer module, GRU, and the multi-head temporal convolution layer further enhance accuracy. 3) Comparing “w/o Glstm2D” and “w/o Simformer” highlights that, while grid cell features (Simformer) are helpful, area demand change features (Glstm2D) are more critical for accurate prediction. This justifies the re-integration of Glstm2D in the feature fusion stage (Section 4.3.1).

To validate our fusion strategy in Glstm2D module, we conducted comparison experiments of summation, gated, attention based, and concatenation methods. The results reveal that concatenation achieves a superior balance of performance across all metrics. We attribute this to three factors. First, concatenation fully preserves the forward and backward features for downstream processing, unlike weighted sum mechanisms which risk premature information compression. Second, given the high correlation observed between the forward and backward hidden states, complex adaptive weighting schemes offer marginal benefits and may introduce noise or overfitting. Finally, the powerful reweighting capabilities of our downstream Simformer and FF-TM modules render an additional adaptive fusion stage redundant. These findings strongly justify our selection of concatenation as the final strategy.

5.7. Visualization of OD demand prediction

To further demonstrate the effectiveness of DSTCN, we visualize its OD demand prediction results from two perspectives.

Overall matrix comparison: To directly assess model accuracy at the network scale, we compare the predicted OD demand matrices against ground truth. As MPGCN and CMOD are strong OD baselines, they are included for comparison. Fig. 8(a) presents the actual demand matrix, while the corresponding predictions from DSTCN, MPGCN, and CMOD are shown in the first row. Visually, DSTCN’s predictions most closely match the true matrix. For finer-grained analysis, we zoom in on

Table 9

Ablation study results on NYC-TOD2018 and NYC-TOD2019.

Variant	NYC-TOD2018		NYC-TOD2019	
	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓
w/o D-BiLSTM	1.490	3.323	1.296	2.944
w/o D-STGL	1.473	3.303	1.277	2.915
w/o O-BiLSTM	1.487	3.291	1.286	2.929
w/o O-STGL	1.478	3.294	1.279	2.924
w/o Glstm2D	1.533	3.364	1.314	2.959
w/o External	1.489	3.356	1.286	2.946
w/o Simformer	1.484	3.318	1.280	2.928
w/o GRU	1.548	3.379	1.319	2.952
w/o Mul-TCL	1.474	3.300	1.269	2.914
DSTCN (Sum)	1.489	3.318	1.283	2.952
DSTCN (Attention)	1.477	3.322	1.269	2.913
DSTCN (Gated)	1.475	3.316	1.274	2.926
DSTCN	1.469	3.284	1.273	2.909

a selected region of the matrix (second row). DSTCN consistently provides predictions nearest to ground truth, whereas MPGCN and CMOD display larger errors in certain cells. These results confirm that DSTCN accurately captures the spatial and temporal patterns of OD demand.

Temporal demand fluctuation: We further evaluate model performance by selecting three high-demand OD pairs and plotting predictions over 10 consecutive days (December 22–31, 2018). As shown in Fig. 9, actual OD demand fluctuates significantly, with pronounced daily peaks. DSTCN’s predictions closely track these trends, capturing both intra-day and day-to-day variations, whereas baselines show more pronounced deviations. This indicates DSTCN’s robustness in modeling complex, time-varying OD demand.

5.8. Implementation details analysis

We conduct sensitivity analysis on DSTCN’s major hyperparameters using the HZMetro dataset (Fig. 10). For each ablation, all other parameters are fixed to their default values described in Section 5.1.2.

Feature dimension D . As shown in Fig. 10(a), increasing D up to 256 improves MAE and RMSE by enhancing the model’s expressive power. However, a further increase to $D = 512$ leads to overfitting and higher errors. Thus, $D = 256$ is selected for optimal performance.

GRU feature dimension. Fig. 10(b) demonstrates that performance improves as the GRU feature dimension increases, with the best results at 512. Beyond this, MAE rises sharply due to overfitting, so 512 is adopted as the optimal setting.

Input period ratios. We explore different ratios for weekly, daily, and recent input periods in Fig. 10(c). While a 2:2:2 split minimizes MAE, a 2:1:3 split yields the lowest RMSE. Overall, recent and weekly inputs provide more useful information than daily inputs for OD prediction.

Number of simformer attention heads. Finally, as shown in Fig. 10(d), changing the number of Simformer heads between 2, 4, 6, and 8 produces minimal effect on prediction performance, indicating DSTCN’s robustness to this hyperparameter.

5.9. Analysis of regional OD demand change trends

To systematically investigate the presence of regional spatio-temporal correlations in OD demand, we randomly select five adjacent regions from the NYC Taxi OD dataset and visualize their demand trajectories over all possible destinations (Fig. 11). The results reveal pronounced synchronous peaks and valleys in demand for specific destinations across these regions, indicating strong temporal and spatial coupling among neighboring origins. Conversely, Fig. 12 examines the inbound demand to these five destinations from all origins, again highlighting consistent variation patterns from adjacent origins.

These empirical findings underscore two critical properties of the OD demand matrix: (1) Demand trajectories from adjacent origins to all des-

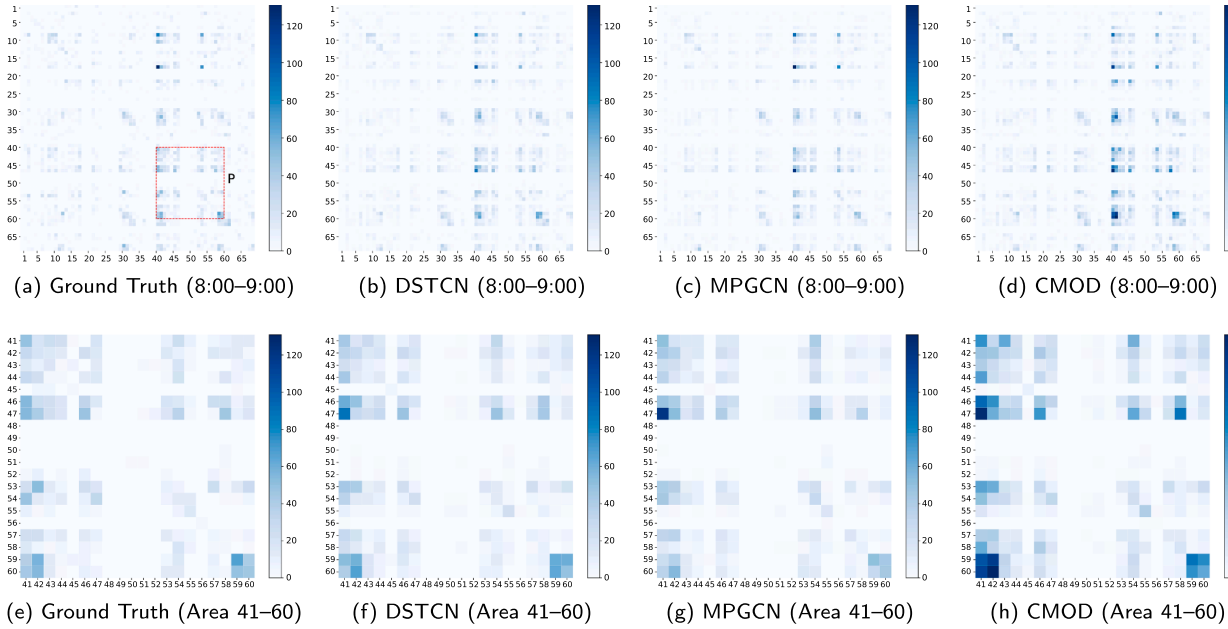
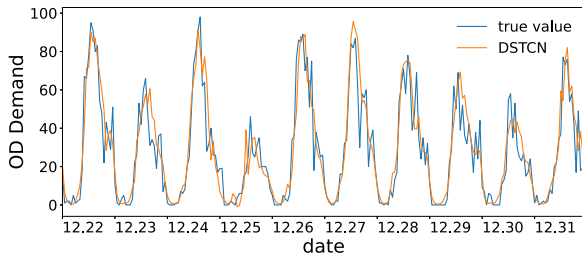
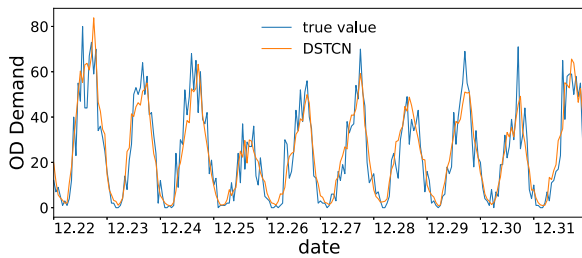


Fig. 8. Prediction results of OD distribution on December 28. The y-axis represents the origin area and the x-axis represents the destination area. (a)–(d): Full grid (8:00–9:00). (e)–(h): Partial grid (areas 41–60).



(a) Demands (area 40 → area 59).



(b) Demands (area 59 → area 68).

Fig. 9. Comparison between OD grid cell prediction and actual value. Demands (area 40 → area 59). Demands (area 59 → area 68).

tinuations tend to be highly correlated; (2) Inbound demand from all origins to adjacent destinations also exhibits substantial similarity. We refer to this phenomenon as the **Regional OD Demand Change Tendency Correlation**. Such bidirectional spatio-temporal regularity directly motivates the explicit dual-axis modeling in our proposed method, aiming to exploit both origin- and destination-driven dependencies for robust OD demand prediction.

5.10. Analysis of temporal similarity among OD demand

To quantitatively assess the dynamic temporal correlation of OD demand across regions, Fig. 13 visualizes the OD demand matrices for ten randomly selected areas in Manhattan over two consecutive time inter-

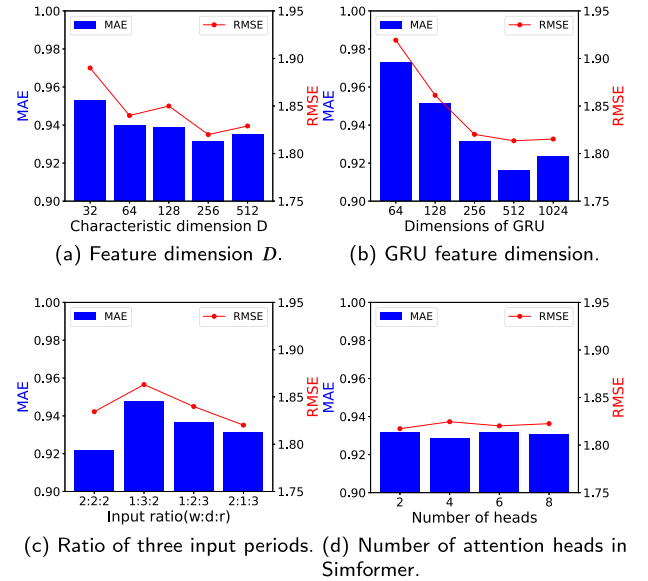


Fig. 10. MAE and RMSE of DSTCN under different hyperparameter settings on HZMetro. (a)Feature dimension D . (b)GRU feature dimension. (c)Ratio of three input periods. (d)Number of attention heads in Simformer.

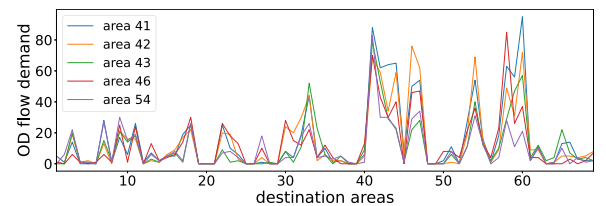


Fig. 11. Visualization of outbound OD demand trajectories from five adjacent origins (areas 41, 42, 43, 46, 54) to all destinations on NYC-TOD2018. X-axis: destination index; Y-axis: demand count. Synchronous patterns indicate strong regional correlation.

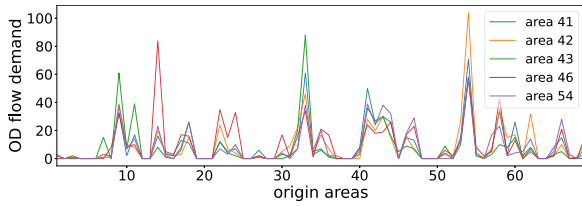


Fig. 12. Inbound OD demand trajectories to the same five regions from all origins on NYC-TOD2019. High similarity across adjacent destinations further confirms bidirectional spatio-temporal correlation.

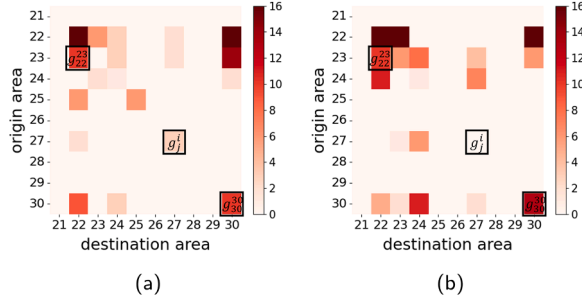


Fig. 13. Temporal snapshots of dynamic similarity among OD grid cells at (a) 8:00–9:00 and (b) 9:00–10:00, highlighting evolving correlation structures in the OD matrix.

vals. In these heatmaps, each grid cell g_i^j denotes the passenger demand from area i to area j , where similar colors indicate close demand magnitudes.

It is notable that some non-adjacent OD pairs, such as g_{22}^{23} and g_{30}^{30} in Fig. 13(a), exhibit analogous demand patterns, highlighting that high demand similarity is not restricted to spatial proximity. Furthermore, comparing Fig. 13(a) and (b) reveals that these demand similarity structures are not static but evolve over time. This underscores the temporally dynamic and complex nature of OD correlations in urban mobility, which motivates the incorporation of temporally adaptive modules in our modeling framework.

6. Conclusions and future work

We propose the Dynamic Spatio-Temporal Correlation Network (DSTCN) for OD demand forecasting. By modeling temporal variation and spatial dependency from both origin and destination perspectives via Glsim2D and Simformer, and adaptively fusing with GRU and multi-head temporal convolution, DSTCN achieves robust and accurate prediction. Experiments on three real-world datasets show superiority over state-of-the-art baselines, with ablations validating module contributions and the benefits of bidirectional temporal modeling. Beyond accuracy, capturing directional and grid-level correlations enhances interpretability and supports traffic management and resource allocation.

Practically, compared with traditional OD flow prediction (Gong et al., 2023, 2021), OD demand data directly reflects the number of trips between regions at each time step, reducing noise from inter-regional movement and improving timeliness for ITS such as fine-grained dispatch and congestion mitigation.

In future work, we will extend DSTCN to support fine-grained demand inference from coarse-grained observations, enhancing its adaptability to complex urban environments and partially observed data. Furthermore, inspired by the success of pre-trained foundation models, a promising direction is to investigate whether the core components of DSTCN can be enhanced by these paradigms. For instance, one could explore initializing our model with weights from a spatio-temporal foundation model and then fine-tuning it on the OD prediction task, potentially improving data efficiency and generalization. This direction will further

empower ITS applications with more accurate, reliable, and scalable OD demand prediction.

CRediT authorship contribution statement

Yongshun Gong: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition; **Piao Yu:** Methodology, Conceptualization, Validation, Data curation, Visualization, Writing – original draft; **Xu Zhang:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing; **Xinxin Zhang:** Investigation, Data curation, Validation, Writing – review & editing; **Xiushan Nie:** Validation, Resources, Writing – review & editing; **Haoliang Sun:** Investigation, Formal analysis, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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